

Functional requirements

1. User registration and profile management

1. The system shall allow older adults to create an account using email, phone number, or caregiver-assisted registration.
2. The system shall allow users to enter basic profile information such as name, age, gender, emergency contacts, medical conditions, allergies, and preferred language.
3. The system shall allow users to update their health profile at any time.
4. The system shall allow users to nominate caregivers, family members, or healthcare professionals who can access selected health information.
5. The system shall support different user roles, including older adult, caregiver, doctor, nurse, and administrator.
6. The system shall allow users to set accessibility preferences such as larger text, high contrast mode, simplified navigation, and voice interaction.

2. Medication management

1. The system shall allow users to add prescribed medications, including dosage, frequency, timing, and special instructions.
2. The system shall send medication reminders at scheduled times.
3. The system shall allow users to confirm whether they have taken, skipped, or delayed a medication.
4. The system shall notify caregivers if a critical medication is repeatedly missed.
5. The system shall provide warnings for potential medication conflicts based on known user allergies or recorded medications.
6. The system shall allow healthcare professionals or caregivers to update medication schedules with user permission.

3. AI health assistant

1. The system shall provide an AI-powered virtual assistant that can answer general health-related questions in simple language.
2. The system shall allow users to interact with the AI assistant using text or voice.
3. The system shall provide health guidance based on the user's profile, symptoms, and history where appropriate.
4. The system shall clearly advise users to contact a healthcare professional or emergency service when symptoms may indicate a serious condition.
5. The system shall not provide a final medical diagnosis without professional clinical review.
6. The system shall explain AI-generated suggestions in plain, understandable terms.

4. Symptom checking and health monitoring

1. The system shall allow users to record symptoms such as pain, dizziness, shortness of breath, fatigue, fever, or confusion.
2. The system shall ask follow-up questions to better understand reported symptoms.
3. The system shall classify symptoms by urgency, such as low, moderate, urgent, or emergency.
4. The system shall recommend next steps, such as self-care, booking an appointment, contacting a caregiver, or calling emergency services.
5. The system shall allow users to record daily health status, mood, sleep, appetite, and physical activity.
6. The system shall detect unusual changes in health patterns and notify the user or caregiver where appropriate.

5. Vital signs and device integration

1. The system shall allow users to manually enter vital signs such as blood pressure, heart rate, blood glucose, oxygen saturation, temperature, and weight.
2. The system shall integrate with compatible wearable devices or home monitoring devices.
3. The system shall display health trends over time using simple charts and summaries.
4. The system shall alert users when recorded values fall outside safe or personalised ranges.
5. The system shall allow healthcare providers to review shared vital sign data with user consent.

6. Appointment management

1. The system shall allow users to record upcoming medical appointments.
2. The system shall send reminders before scheduled appointments.
3. The system shall allow users to store appointment details such as doctor name, clinic location, reason for visit, and preparation notes.
4. The system shall allow caregivers to view or manage appointments with user permission.
5. The system shall support telehealth appointment links where available.
6. The system shall allow users to record questions they want to ask their doctor.

7. Emergency support

1. The system shall provide a clearly visible emergency assistance button.
2. The system shall allow users to contact emergency services from within the app.
3. The system shall allow users to notify emergency contacts during a health emergency.
4. The system shall share critical medical information with emergency contacts if the user has enabled this feature.
5. The system shall detect emergency keywords or severe symptoms and recommend urgent action.
6. The system shall provide location-sharing support during emergencies, subject to user consent and device permissions.

8. Caregiver and family support

1. The system shall allow users to invite trusted caregivers or family members.
2. The system shall allow users to control what information caregivers can see.
3. The system shall notify caregivers about missed medication, abnormal health readings, emergency alerts, or major behaviour changes.
4. The system shall allow caregivers to send reminders or supportive messages to the older adult.
5. The system shall allow caregivers to view health summaries and trends.
6. The system shall maintain an activity log of caregiver actions for transparency.

9. Health education

1. The system shall provide educational content on healthy ageing, fall prevention, nutrition, exercise, chronic disease management, medication safety, and mental wellbeing.
2. The system shall personalise educational content based on user profile and health goals.
3. The system shall present educational material in simple language with large text and optional audio playback.
4. The system shall allow users to save helpful articles or videos.
5. The system shall provide reminders for healthy habits such as hydration, walking, stretching, and sleep routines.

10. Mental health and social wellbeing

1. The system shall allow users to record mood and emotional wellbeing.
2. The system shall detect signs of loneliness, anxiety, or low mood based on user inputs and interaction patterns.
3. The system shall suggest supportive actions such as contacting a friend, doing a breathing exercise, or speaking with a professional.
4. The system shall provide access to mental health helplines or support resources.
5. The system shall allow caregivers to receive alerts if serious wellbeing concerns are detected, with user consent.

11. Personalised recommendations

1. The system shall generate personalised health reminders based on the user's age, health conditions, medication schedule, and activity patterns.
2. The system shall recommend daily wellness activities suited to the user's mobility and health status.
3. The system shall adapt recommendations based on user feedback and past behaviour.
4. The system shall avoid recommendations that conflict with known medical restrictions.
5. The system shall clearly indicate when recommendations are AI-generated.

12. Notifications and reminders

1. The system shall send notifications for medication, appointments, hydration, exercise, health checks, and follow-up tasks.
2. The system shall allow users to customise notification frequency and style.
3. The system shall support voice reminders for users with visual difficulties.
4. The system shall repeat critical reminders if no response is received.
5. The system shall avoid excessive notifications to reduce confusion or alarm.

13. Reports and health summaries

1. The system shall generate daily, weekly, or monthly health summaries.
2. The system shall allow users to export health reports for healthcare appointments.
3. The system shall provide simple explanations of health trends.
4. The system shall allow caregivers and doctors to access reports only with user permission.
5. The system shall highlight important changes, missed medications, abnormal readings, or recurring symptoms.

14. Feedback and support

1. The system shall allow users to provide feedback about app usability and AI responses.
 2. The system shall allow users to report incorrect or unsafe AI advice.
 3. The system shall provide access to technical support.
 4. The system shall include a help section with step-by-step instructions.
 5. The system shall allow users to request human assistance where available.
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Non-functional requirements

1. Usability

1. The app shall use a simple and intuitive interface suitable for older adults.
2. The app shall minimise the number of steps required to complete common tasks.
3. The app shall use clear labels, large buttons, and plain language.
4. The app shall avoid technical medical jargon where possible.
5. The app shall provide confirmation messages after important actions, such as medication logging or emergency contact updates.
6. The app shall support guided onboarding for first-time users.
7. The app shall be usable by people with limited digital literacy.

2. Accessibility

1. The app shall comply with recognised accessibility guidelines such as WCAG 2.2 where applicable.
2. The app shall support large text sizes.

3. The app shall support screen readers.
4. The app shall provide high contrast display options.
5. The app shall support voice input and audio output.
6. The app shall avoid relying solely on colour to communicate important information.
7. The app shall support users with reduced vision, hearing, dexterity, or cognitive ability.
8. The app shall allow users to slow down or repeat voice instructions.

3. Safety

1. The app shall clearly state that AI advice does not replace professional medical advice.
2. The app shall escalate potentially serious symptoms to emergency or professional care guidance.
3. The app shall avoid providing unsafe, overconfident, or unsupported medical recommendations.
4. The app shall include safeguards against hallucinated medical advice.
5. The app shall provide clear emergency instructions when urgent symptoms are detected.
6. The app shall log AI-generated advice for auditing and safety review.
7. The app shall prioritise user safety over engagement or convenience.

4. Privacy

1. The app shall collect only the health data necessary for its intended functions.
2. The app shall obtain user consent before collecting, storing, or sharing health information.
3. The app shall allow users to view, edit, export, or delete their personal data.
4. The app shall allow users to control which caregivers or clinicians can access their information.
5. The app shall not share personal health data with third parties without explicit consent.
6. The app shall anonymise or de-identify data where used for analytics or model improvement.
7. The app shall clearly explain how user data is used.

5. Security

1. The app shall encrypt sensitive data in transit and at rest.
2. The app shall support secure authentication, such as passwords, PINs, biometrics, or multi-factor authentication.
3. The app shall automatically log users out after a period of inactivity where appropriate.
4. The app shall protect against unauthorised access to health records.
5. The app shall maintain audit logs for access to sensitive information.

6. The app shall use role-based access control for older adults, caregivers, clinicians, and administrators.
7. The app shall regularly undergo security testing and vulnerability assessments.

6. Reliability

1. The app shall be available whenever users need to access critical health features.
2. The app shall continue to provide basic reminders even during temporary network issues.
3. The app shall recover gracefully from system errors.
4. The app shall prevent data loss during crashes or connectivity interruptions.
5. The app shall clearly inform users when a feature is unavailable.
6. The app shall maintain accurate reminder scheduling across time zones and daylight saving changes.

7. Performance

1. The app shall load key screens within a reasonable time on older smartphones.
2. The app shall respond quickly to user interactions.
3. The AI assistant shall provide responses within an acceptable timeframe.
4. The app shall minimise battery usage, especially when connected to wearable devices.
5. The app shall operate efficiently on devices with limited processing power or memory.
6. The app shall support offline access to essential information such as medication schedules and emergency contacts.

8. Accuracy

1. The app shall provide health recommendations based on reliable medical knowledge sources.
2. The app shall distinguish between confirmed user data and AI-inferred information.
3. The app shall avoid presenting uncertain AI outputs as facts.
4. The app shall provide confidence levels or uncertainty indicators where appropriate.
5. The app shall regularly update its medical knowledge base.
6. The app shall allow clinical review of high-risk AI outputs.
7. The app shall correct or remove inaccurate recommendations once identified.

9. Maintainability

1. The system shall be designed using modular components.
2. The system shall allow updates to AI models without disrupting core app functions.
3. The system shall support regular software updates and bug fixes.
4. The system shall maintain clear documentation for developers and administrators.

5. The system shall allow new health monitoring devices or integrations to be added in the future.
6. The system shall support testing of individual modules such as reminders, AI assistant, notifications, and data sharing.

10. Scalability

1. The system shall support an increasing number of users without major performance degradation.
2. The system shall support scaling of AI services during peak usage.
3. The system shall support large volumes of health data from users and connected devices.
4. The system shall allow expansion to additional languages, regions, and healthcare providers.
5. The system shall support future integration with hospitals, pharmacies, and electronic health record systems.

11. Interoperability

1. The app shall support integration with common wearable devices and health monitoring tools.
2. The app shall support standard healthcare data formats where applicable.
3. The app shall allow health data export in commonly used formats such as PDF or CSV.
4. The app shall support integration with telehealth platforms.
5. The app shall support future integration with electronic health record systems, subject to privacy and security requirements.

12. Ethical requirements

1. The app shall avoid ageist assumptions in its design and recommendations.
2. The app shall respect user autonomy and allow older adults to make their own decisions where possible.
3. The app shall clearly distinguish between AI advice and human clinical advice.
4. The app shall avoid manipulating users into unnecessary engagement.
5. The app shall be designed to support independence, not replace human care.
6. The app shall provide transparency about how AI decisions are made.
7. The app shall include processes for reviewing bias in AI recommendations.

13. Compliance

1. The app shall comply with relevant privacy and health data protection laws in the target region.
2. The app shall comply with medical software regulations if its features qualify as medical device functionality.
3. The app shall maintain records needed for clinical safety audits.

4. The app shall provide consent management features.
5. The app shall support data retention and deletion policies.
6. The app shall provide users with clear terms of use and privacy notices.

14. Availability and backup

1. The system shall maintain regular backups of user data.
2. The system shall support disaster recovery procedures.
3. The system shall minimise downtime for critical services.
4. The system shall notify users of planned maintenance where possible.
5. The system shall ensure emergency information remains accessible during service disruption.

15. User trust and transparency

1. The app shall explain why important alerts or recommendations were generated.
2. The app shall show users when data was last updated.
3. The app shall identify whether advice came from AI, a caregiver, or a healthcare professional.
4. The app shall allow users to challenge, correct, or dismiss AI-generated suggestions.
5. The app shall avoid creating false reassurance when symptoms may require medical attention.
6. The app shall provide clear disclaimers without overwhelming users.